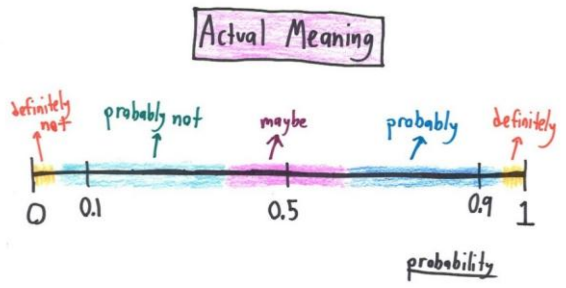
EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 7 - Lecture No 13 – February 23, 2026



Announcements!

1. Office hours (Ramadan) – Tuesday & Thursday 11:30 to 12:30 PM @ Room C-211 (or through meeting request)
2. Quiz No 04 (on DRV) - February 25, 2026
3. Midterm Examination - March 3, 2026, from 03:25 PM to 04:55 PM at the Tariq Rafi Lecture Hall
   - ✓ 7 days to go!
   - ✓ Practice Problems uploaded & Midterm Exam for Fall 2025 uploaded
4. Assignment 2 – due March 8, 2026
5. TA's EHSAS Hours

EHSAS TUTORS' RAMADAN SCHEDULE - Spring 2026							
Course	Tutor	Email	Monday	Tuesday	Wednesday	Thursday	Friday
Intro to Probability & Statistics	Rayyan Ali Shah	rs08770@st.habib.edu.pk	2:00-3:00		2:00-3:00		12:40-1:40
	Hiba Aiman Ali	ha09269@st.habib.edu.pk	12:45-2:30		12:45-2:30		11:00-12:00
	Zain Naqi	zn09224@st.habib.edu.pk		9:00 - 10:30	11:45 - 12:45	9:00 - 10:30	

Agenda for today

1. Unit 4 – Discrete Random Variables

Unit 4
Discrete Random Variables

TAs -

EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics, Spring 2026

Meet Your TAs
Zain Naqi

About Me
I am a Computer Science Junior. My career interests include machine learning, deep learning, and reinforcement learning. Apart from academics, I do competitive programming and run a small e-commerce business. Mostly, you will find me in the library discussion area.

Ehsaas Hours
- Tue 9:30 - 11:30
- Thu 9:30 - 11:30

Contact Me
zn09224@st.habib.edu.pk or “Zain Naqi” on Teams

Meet Your TAs
Rayyan Shah

About Me
I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours
- Mon 4:00-5:00
- Wed 4:00-5:00
- Fri 3:30-4:30

Contact Me
rs08770@st.habib.edu.pk
Or “Rayyan Ali Shah” on Teams

Meet Your TAs
Hiba Aiman Ali

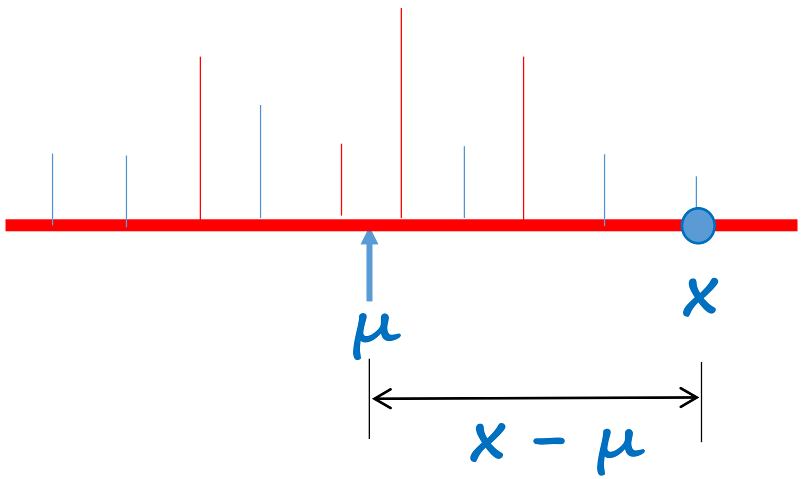
About Me
I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I’m not in class, I’m most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours
- Mon 2:30 - 4:00
- Wed 2:30 - 4:00
- Fri 12:00-1:00

Contact Me
ha09269@st.habib.edu.pk on Teams or Outlook

Variance – a measure of the spread of a PMF

2 r.v's Y & X with same mean μ



- Random variable X, with mean $\mu = E[X]$
- Distance from the mean: $X - \mu$?
- Average distance from the mean?

$$E[X - \mu] = E[X] - \mu = \mu - \mu = 0$$

- **Definition of variance:** $var(X) = E[(X - \mu)^2]$

≥ 0

$g(x) = (x - \mu)^2$

- Calculation, using the expected value rule, $E[g(X)] = \sum_x g(x) p_X(x)$

$$var(X) = E[g(X)] = \sum_x (x - \mu)^2 p_X(x)$$

- **Standard deviation:** $\sigma_X = \sqrt{var(X)}$

Properties of the variance

$$var(aX + b) = a^2 var(X)$$

$var(3 - 4x)?$
$= (-4)^2 var(X)$
$= 16\, var(X)$

- Notation: $\mu = \boldsymbol{E}[\boldsymbol{X}]$
- Let Y = X + b

$\gamma = E[Y] = \mu + b$

$var(Y) = E\left[(Y - \gamma)^2\right] = E[(X + \cancel{b} - (\mu + \cancel{b}))^2] = E\left[(X - \mu)^2\right] = var(X)$

- Let Y = aX

$\gamma = E[Y] = a\mu$

$var(Y) = E\left[(aX - a\mu)^2\right] = E\left[a^2 (X - \mu)^2\right] = a^2 E\left[(X - \mu)^2\right] = a^2 var(X)$

A useful formula: $var(X) = E[X^2] - (E[X])^2$

$var(X) = E\left[(X - \mu)^2\right] = E\left[X^2 - 2\mu X + \mu^2\right]$

$= E\left[X^2\right] - 2\mu E[X] + \mu^2 = E\left[X^2\right] - (E[X])^2$

Variance of Bernoulli r.v

A useful formula: $var(X) = E[X^2] - (E[X])^2$

$$X = \begin{cases} 1, & \text{w.p.} \quad p \\ 0, & \text{w.p.} \quad 1-p \end{cases}$$

$E[X] = 1(p) + 0(1 - p) = p$

$$X^2 = \begin{cases} 1, & \text{w.p.} \quad p \\ 0, & \text{w.p.} \quad 1-p \end{cases}$$

$E[X^2] = 1(p) + 0(1 - p) = p$

$$\begin{aligned} var(X) &= E[X^2] - (E[X])^2 \\ &= p - p^2 \\ &= p\,(1-p) \end{aligned}$$

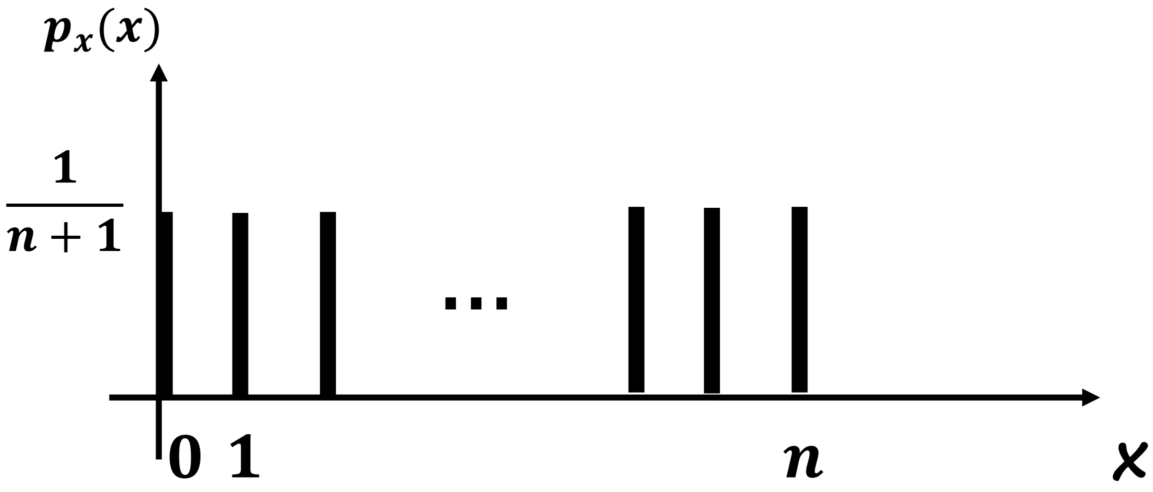
Exercise –

i. Plot Var(X) for different possible values of p.

ii. What does the plot tell you?

iii. What does the plot tell you about the fair coin?

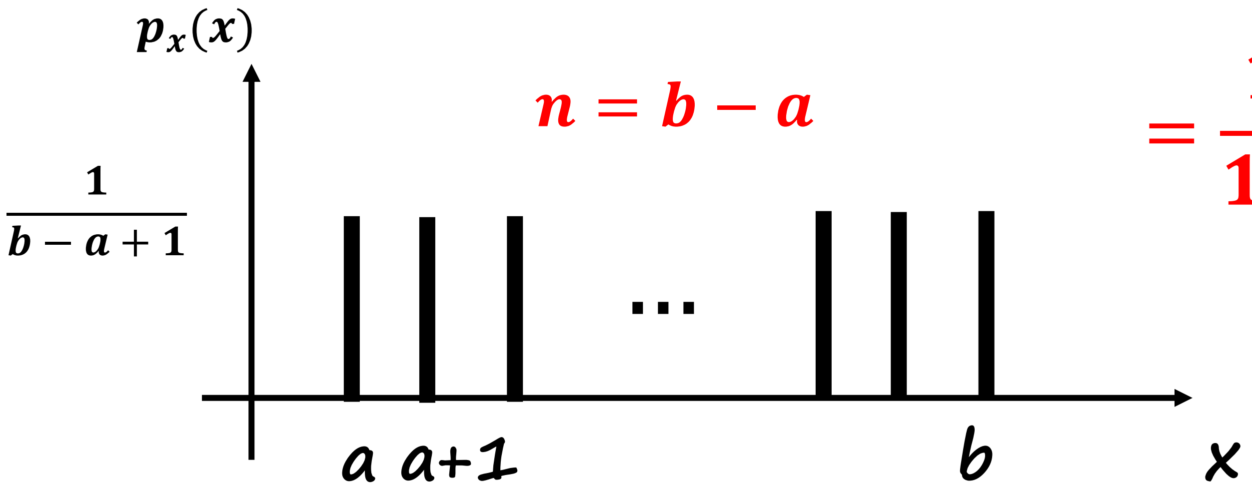
Variance of the uniform



$$var(X)=E\left[X^2\right]-(E[X])^2=\frac{1}{n+1}\left(\overbrace{0^2+1^2+2^2+\ \dots\ +n^2}^{\frac{1}{6}n(n+1)(2n+1)}\right)-\left(\frac{n}{2}\right)^2$$

$$=\frac{1}{12}n(n+2)$$

$$n=b-a$$



$$Var(x)=\frac{1}{12}(b-a)(b-a+2)$$

Conditional PMF and expectation, given an event

- Condition on an event A → use conditional probabilities

$$p_X(x) = P\,(X = x)$$

$$\underline{p_{X|A}}(x) = P(\underline{X = x|A})$$

assume $P(A) > 0$

$$\sum_x p_X(x) = 1$$

$$\sum_x p_{X|A}(x) = 1$$

$$E[X] = \sum_x x p_X(x)$$

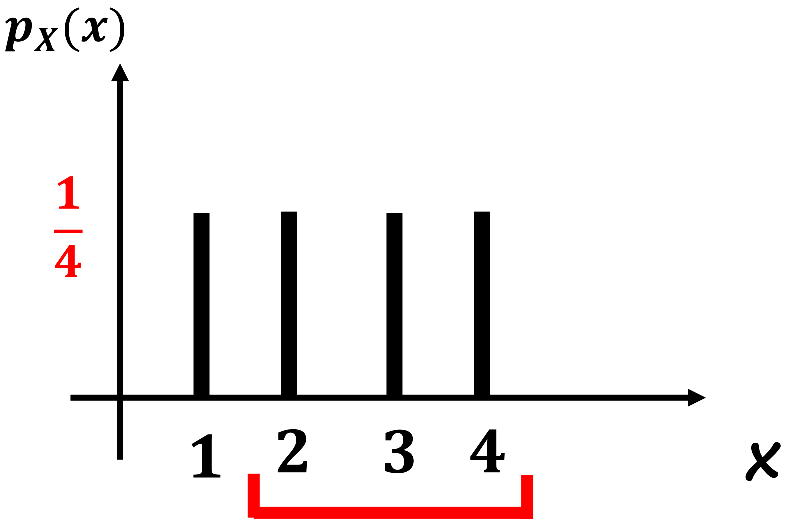
$$E[X|A] = \quad ?$$

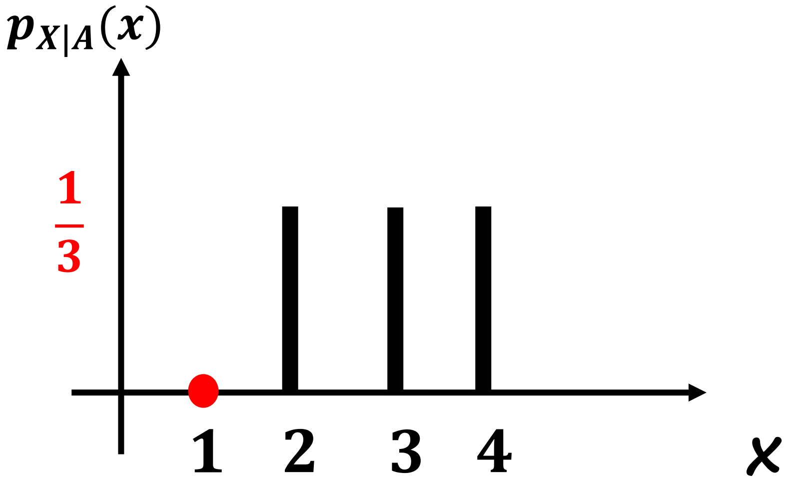
$$E[g(X)] = \sum_x g(x) p_X(x)$$

$$E[g(X)|A] = \sum_x g(x) p_{X|A}(x)$$

Example of conditioning

- Let A = $\{X \geq 2\}$





$E[X] =$ 2.5

$E[X|A] =$ 3

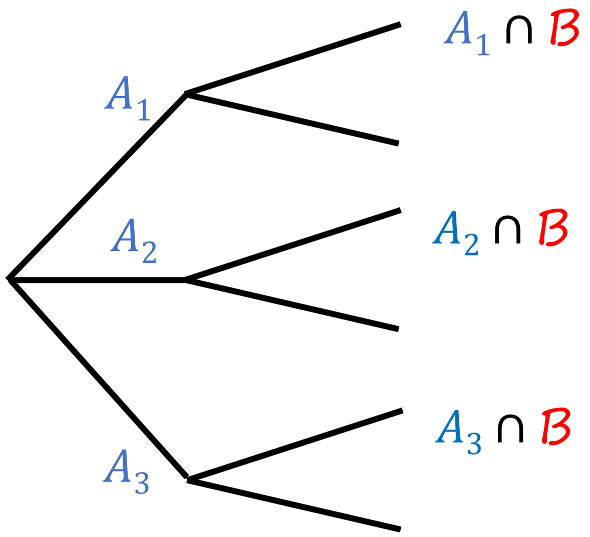
$\text{var}[X] = \frac{1}{12}(b-a)(b-a+2)$

$= \frac{1}{12} * 3 * 5 = \frac{5}{4}$

$\text{var}[X|A] =$

$\frac{1}{3}(4-3)^2 + \frac{1}{3}(3-3)^2 + \frac{1}{3}(2-3)^2 = \frac{2}{3}$

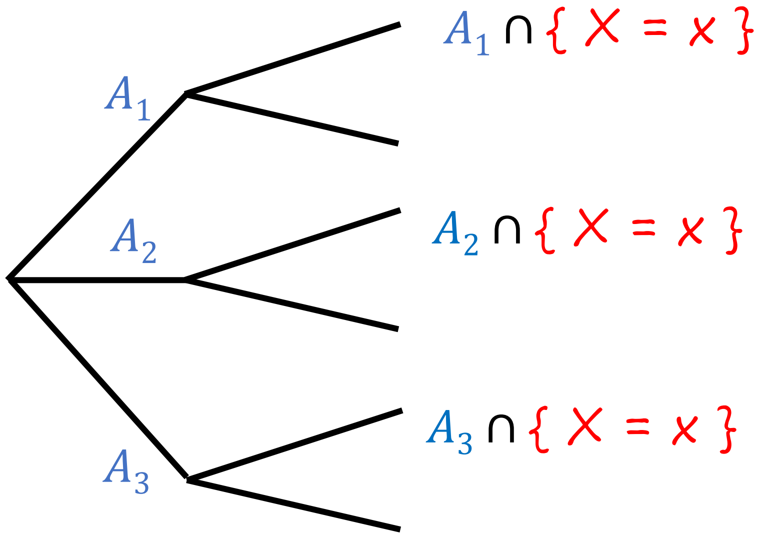
Total Expectation Theorem



$$\mathbf{P}(B) = \mathbf{P}(A_1)\,\mathbf{P}(B \mid A_1) + \dots + \mathbf{P}(An)\,\mathbf{P}(B \mid An)$$

$$B = \{ X = x \}$$

Total Expectation Theorem

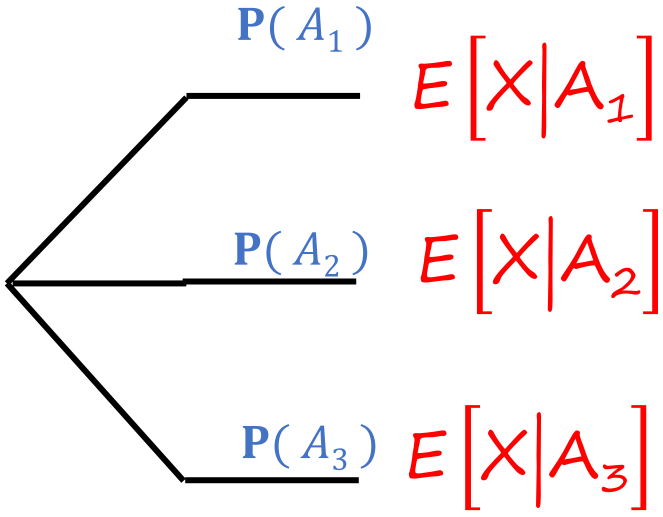


$$\mathbf{P}(B) = \mathbf{P}(A_1)\,\mathbf{P}(B \mid A_1) + \dots + \mathbf{P}(An)\,\mathbf{P}(B \mid An)$$

$$B = \{ X = x \}$$

$$p_X(x) = \mathbf{P}(A_1)\,p_{X|A_1}(x) + \dots + \mathbf{P}(An)\,p_{X|A_n}(x)$$

for all x



$$= \underbrace{\sum_x x\, p_X(x)}_{E[X]} = \mathbf{P}(A_1) \underbrace{\sum_x x\, p_{X|A_1}(x)}_{E[X|A_1]} + \dots$$

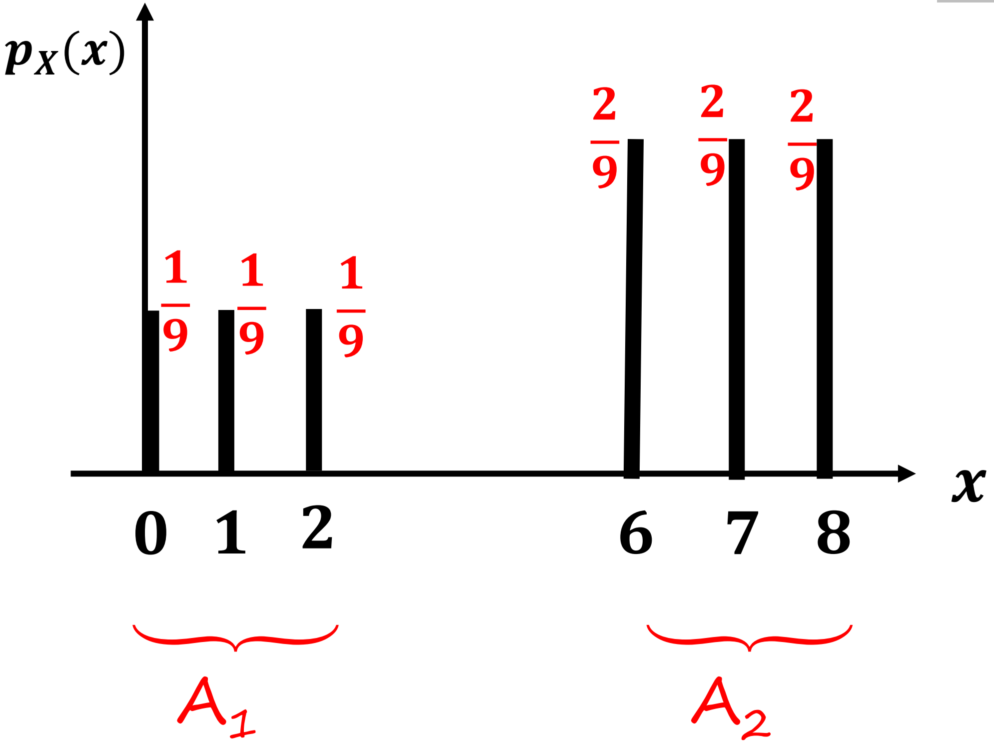
$$\boxed{E[X] = \mathbf{P}(A_1)\,E[X \mid A_1] + \dots + \mathbf{P}(A_n)\,E[X \mid An]}$$

Total Expectation Theorem

Total Expectation Example

Exercise No 01!

Using Total expectation theorem, Calculate the E[X]



➡ $P(A_1) = 1/3$

➡ $P(A_2) = 2/3$

➡ $E[\,X \mid A_1\,] = 1$

➡ $E[\,X \mid A_2\,] = 7$

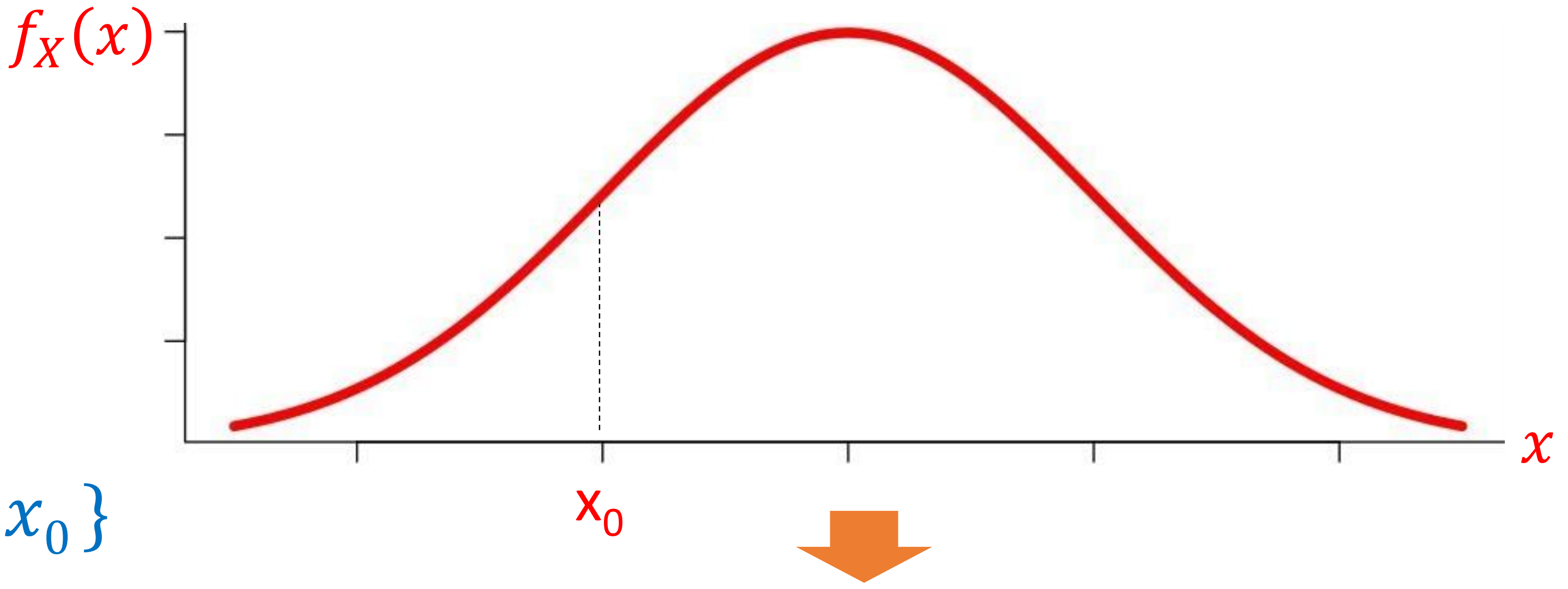
➡ $E[\,X\,] = \frac{1}{3} * 1 + \frac{2}{3} * 7$

$$E[\,X\,] = \mathrm{P}(\mathrm{A}_1)\, E[\,X \mid A_1\,] + \ldots + \mathrm{P}(\mathrm{A}_n)\, E[\,X \mid An\,]$$

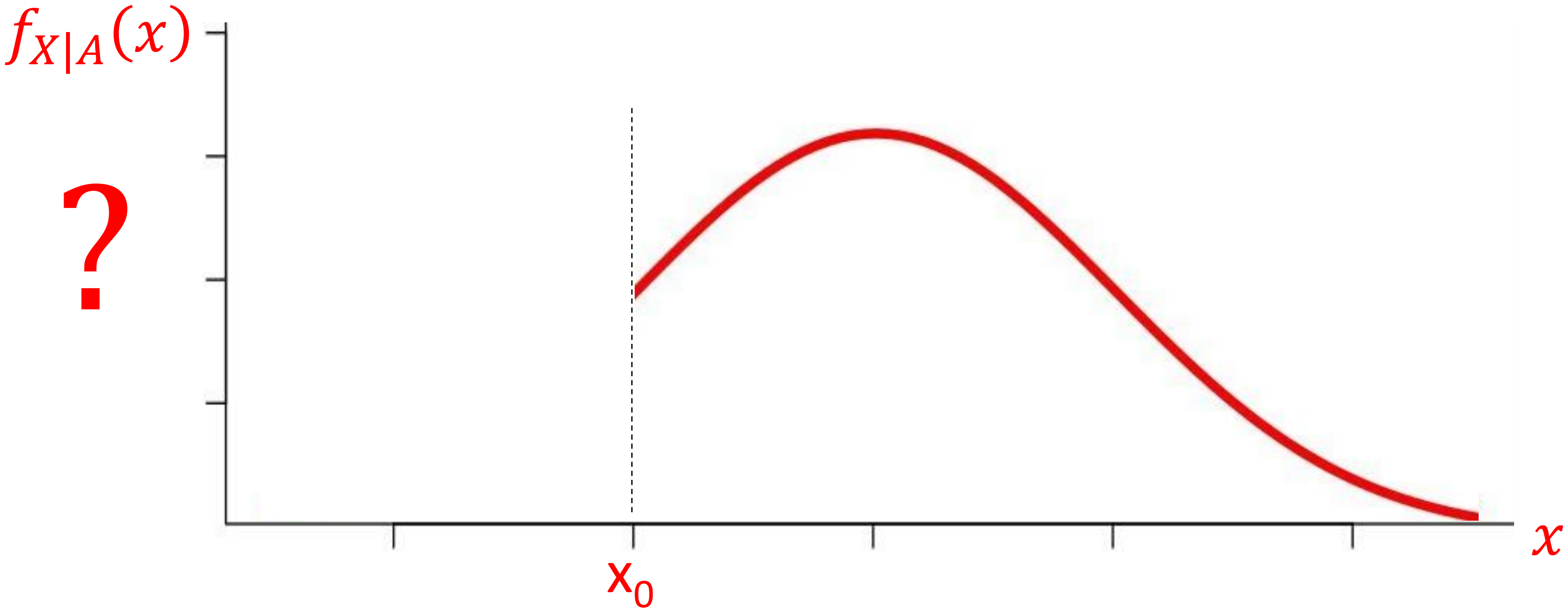
Total Expectation Theorem

Re-visiting the great question by a former student!

"Does the nature of distribution after conditioning remains the same?"

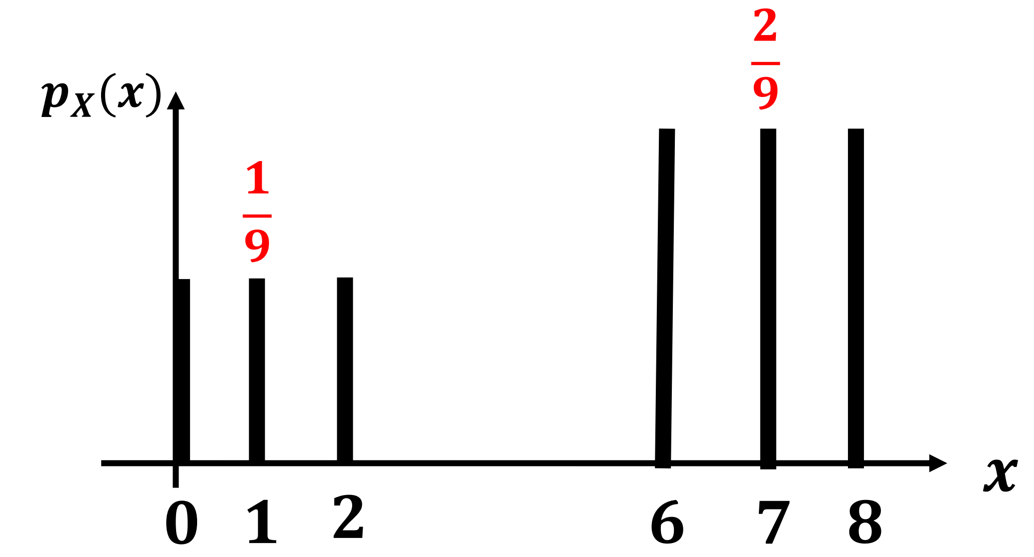


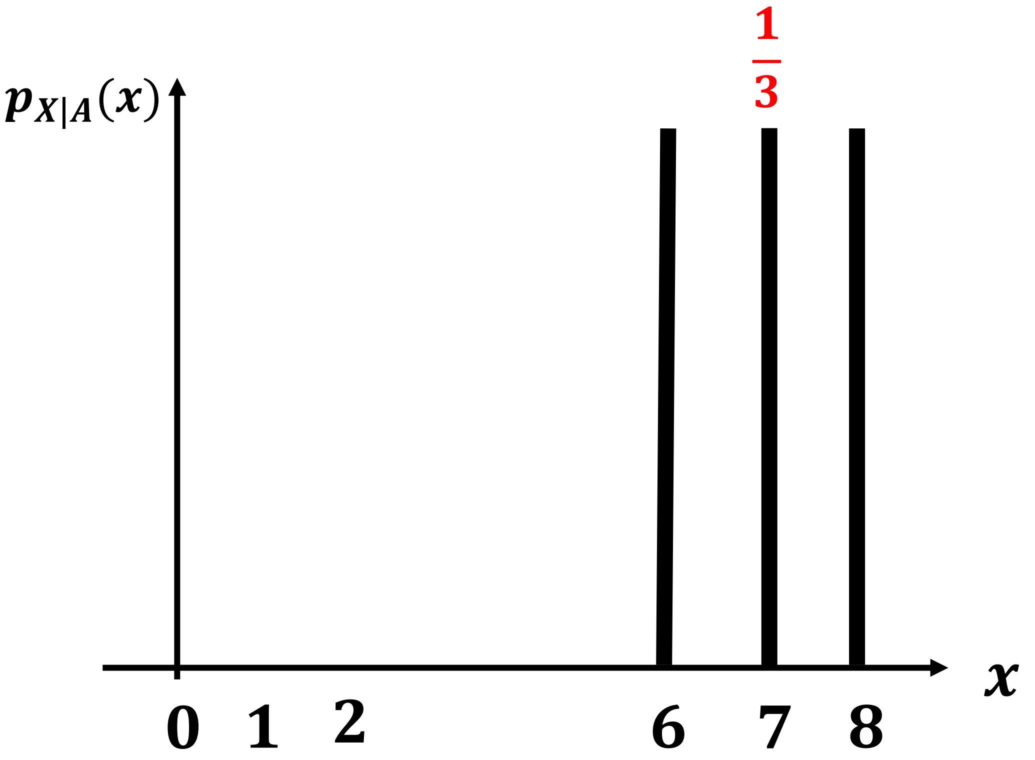
$A = \{ x > x_0 \}$



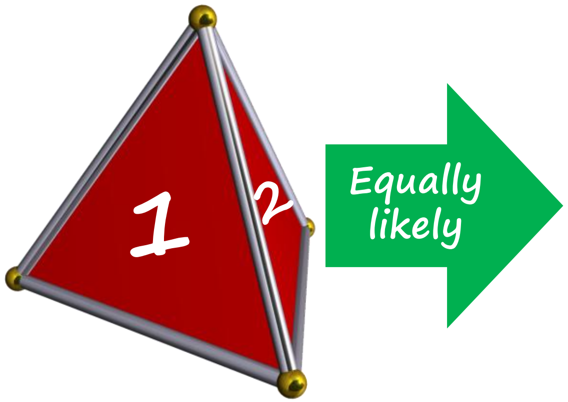
$A = \{ x > 2 \}$

$P_{X|A}(x)$?





Exercise



Problem 3 (25 points): A Random Variable Z_n is defined as $Z_n = max\,(X_1, X_2, ..., X_n)$, where X_i's are outcomes of identical and independently distrubted tetrahedral die (4-sided). As an example $Z_2 = max\,(X_1, X_2)$ - meaning Z_2 is the maximum of the outcome of two independent throws of tetrahedral die X_1 and X_2.

Part a (5 points): Calculate probability $P(Z_{10} < 4)$.

Part b (5 points): Calculate probability $P(Z_3 = 2)$.

Part c (5 points): Calculate and plot the probability mass function, PMF of Z_2.

Part d (10 points): Find the Expected value, $E\,[Z_2|A]$, where event $A = \{X_1 = 3\}$.

Part a

$$Z_{10} < 4 \Rightarrow \max(X_1, \dots X_{10}) < 4)$$

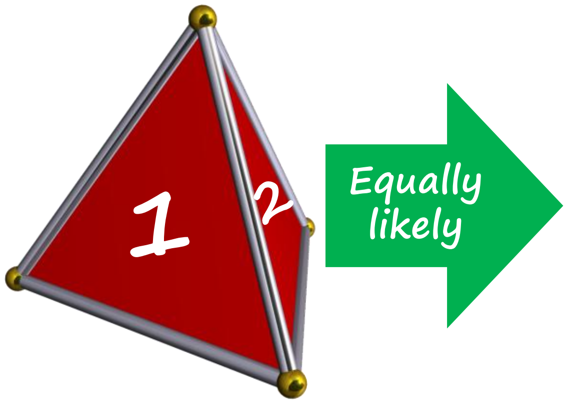
means zero No' of 4 in 10 throws

Probability of No 4 in One throw $= 1 - 1/4$

$= 3/4$

So $P(Z_{10} < 4) = (3/4)^{10} = 0.0563$

Exercise



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Part b

$$P(Z_3 = 2)$$

$$= max(X_1, X_2, X_3) = 2$$

$\Downarrow$

for $Z_3 = 2$

outcomes
1, 1, 2
1, 2, 1
2, 1, 1
2, 2, 1
2, 1, 2
1, 2, 2
2, 2, 2

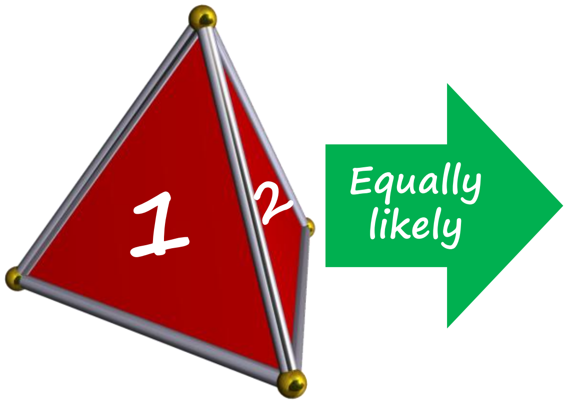
Each Seq probability $= \frac{1}{4^3}$

1	2	2
2	1	2
2	2	1

$$= 7 \times \frac{1}{4^3}$$

$$= 0.109$$

Exercise



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Part c

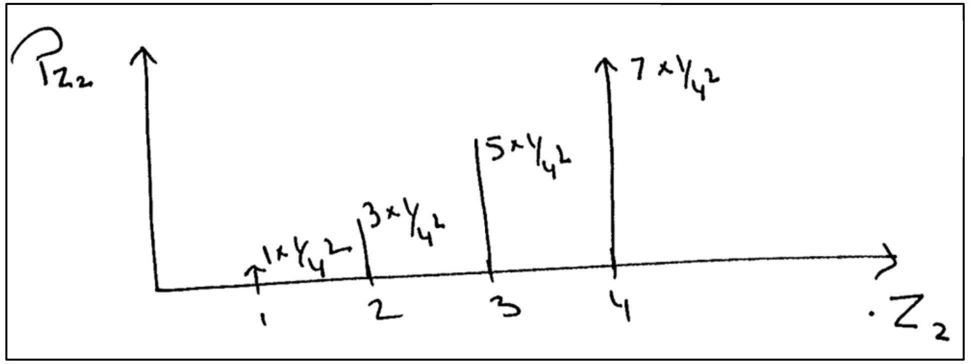
$X_1 \quad X_2$

$Z_2 = 1 \Rightarrow$ 1 1 $\quad 1 \times \frac{1}{4^2}$

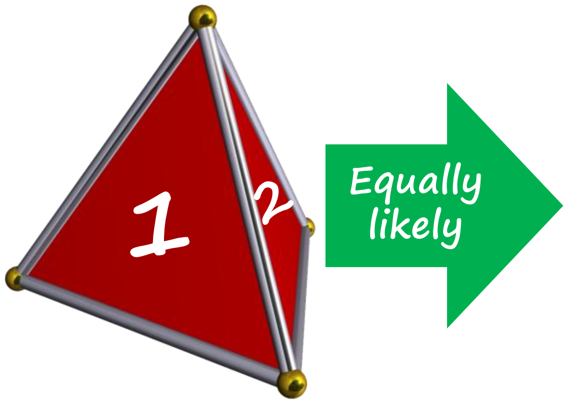
$Z_2 = 2 \Rightarrow$ 1 2, 2 1, 2 2 $\quad 3 \times 1/4^2$

$Z_2 = 3 \Rightarrow$ 1 3, 2 3, 3 3, 3 1, 3 2 $\quad 5 \times 1/4^2$

$Z_2 = 4 \Rightarrow$ 1 4, 2 4, 3 4, 4 4, 4 1, 4 2, 4 3 $\quad 7 \times 1/4^2$



Exercise



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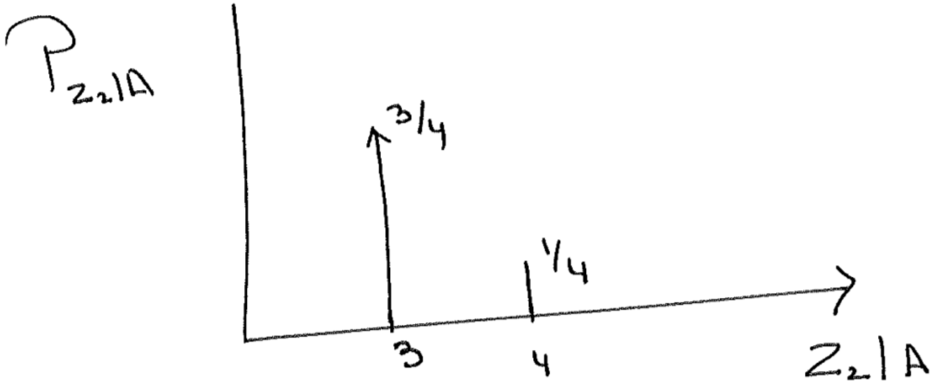
Part d

$$E\left[Z_2 \,|\, (X_1 = 3)\right]$$

X_1	X_2	$P(X_2 = x)$	Z_2
3	1	$1/4$	3
3	2	$1/4$	3
3	3	$1/4$	3
3	4	$1/4$	4



$$E[Z_2|A] = 3 \times 3/4 + 4 \times 1/4$$
$$= \frac{9}{4} + \frac{4}{4} = 13/4$$